

TIYASHA SEN

Denver, CO

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EDUCATION

Northeastern University – College of Engineering, Boston, MA

August 2024

Master of Science in Information Systems

Course Highlights: Program Structures and Algorithms, Application Engineering and Development, Web Design, Network Structures & Cloud Computing, Object Oriented Design, Database Design, Design Patterns

West Bengal University of Technology, Kolkata, India

August 2016

Bachelors, Computer Application

Course Highlights: Algorithms, Web Development, Object Oriented Programming

TECHNICAL SKILLS

Languages: Java, Python, C#, PL/SQL, HTML5

Web Technologies: React.js, Next.js, Node.js, Angular.js, CSS, WordPress

DevOps: AWS, GCP, Pub/Sub, Terraform, SNS, SQS, S3, Step Functions, DynamoDB, IAM, CloudWatch, AWS Lambda, CI/CD, Packer

Frameworks: Spring Boot, Spring MVC, XML, JPA, JSON, MS Office

Database: PostgreSQL, MySQL, Oracle 11g/12c, Cassandra, MongoDB

Tools & Services: Maven, REST, WebLogic, Linux, Shell, Spark, Harness, Postman, JIRA, SOAP UI, SonarQube, Tomcat, JUnit, Eclipse, OAuth2.0

Versioning Tools: Git, SVN

PROFESSIONAL EXPERIENCE

Senior Software Engineer, DISH Network

Sep 2024 – Present

Denver, United States

- Designed and implemented complex workflows using **ASL** within AWS Step Functions and **Java** in **AWS Lambda** optimizing order orchestration and enhancing end-to-end efficiency in order fulfillment processes
- Executed Harness **CI/CD pipelines** adopting to streamline processes, ensuring efficiency on business requirements
- Optimized orchestration workflows to improve system stability by 55% and leveraged **LLMs to build a chatbot**, enhancing automation and infrastructure efficiency.
- Classified defects based on severity, and furnished actionable insights for development, resulting in a weekly cost reduction of 32%
- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline leveraging **AWS OpenSearch** for efficient document retrieval and semantic search, integrating with Nova model to improve answer accuracy and reduce hallucination rates

Student Developer/Senator, Technology Domain of GSG, Northeastern University

Jan 2024 – Aug 2024

Boston, United States

- Constructing scripts in **Python** and Shell on Linux real-time operating system(RTOS) ensuring online presence effectively communicated
- Establishing **AWS CloudFormation** templates for efficient distribution, and used **Terraform** to implement infrastructure as code

Software Developer Intern, Bankers Financial Corporation

Sep 2023 – Jan 2024

Florida, United States

- Administered and boosted the performance of IT infrastructure components with Windows Server OS, to ensure seamless operations
- Fabricated **ASK-B** chatbot using **Java multithreading procedure**, **MySQL** in claims processing workflow improving smooth real-time assistance explaining policy validation rules reducing manual effort by 40%
- Devised **Smart Insertion** script to automate the insertion of data entries, improving data accuracy and processing speed by 40%

Senior Software Developer, Ericsson Global Services Pvt Ltd

Aug 2021 - Aug 2022

Bengaluru, India

- Spearheaded design and development of **Java & Spring Boot** based microservices with SOLID principles utilizing **Cassandra** to optimize network performance metrics, enhancing scalability across telecom infrastructure
- Contributed to **XSD** and **WSDL** schemas to guarantee adherence to industry standards bringing about 50% standardized data consolidation
- Utilized Jenkins for continuous integration and deployment, and **Kafka** for data processing, reducing downtime by 20 hours monthly
- Integrated Google Maps API, Drools, **AWS**(EC2, S3, Lambda) resulting in 85% increase in telecom payments

Software Engineer, Tata Consultancy Services Ltd

Sep 2016 - July 2021

Kolkata, India

- Collaborated on **fine-tuned** query optimization techniques for transaction processing on **Oracle** achieving a 2x secure solutions
- Engineered a **Java, Spring Boot** based integration solution to transfer non-PII customer information from a **legacy credit card** system to a third-party approval platform using RESTful APIs in Agile environment deployed on Weblogic server
- Crafted Angular 7 **security modules** for fraud detection dashboard resulting in reduction of fraudulent transactions by 30%

ACADEMIC PROJECTS

Cloud Native Application | GCP, Terraform, Packer, CI/CD, JWT, REST APIs, Node JS, JUnit

Jan 2024-Apr 2024

Implemented token-based basic authentication and established a CI/CD pipeline reviving GitHub Actions integrated with Terraform and Packer for automated code compilation, testing, deployment, and infrastructure provisioning on Google Cloud Platform (GCP)

Travelling Salesman Problem | Java, Algorithms, Graph Stream

Apr 2023-May 2023

Created solution to optimize paths between crime points from CSV data, applying diverse algorithms- Three-opt, Random swapping, Simulated Annealing, Ant Colony Optimization for graphical representation of optimized TSP tour

BADGES & CERTIFICATIONS

AWS Cloud Quest Badge: Solutions Architect, Cloud Practitioner

2024